

SILVER AS A SAFE-HAVEN

MOTIVATION

- Silver surged 70%+ recently, outperforming gold and major equity indices - raising questions about what drives its returns.
- Growing debate on whether silver is a safe-haven asset or a cyclical, industrial commodity.
- Literature shows silver behaves as a hybrid asset: co-moves with gold but also reacts to oil and equities.
- Understanding silver's daily drivers is crucial for portfolio construction, hedging, and market risk analysis.

RESEARCH QUESTION

To what extent do financial market indicators, other precious metal prices and commodity market conditions explain variations in silver prices, and how do these relationships reveal silver's positioning as a safe-have asset within broader market conditions?

DATA

- Source: Kaggle's Financial Data dataset (FranciscoGCC).
- Relevant: Includes key variables for modeling silver returns.
- Collected: Via Alpha Vantage and FRED (one of the most credible financial market databases)
- Reliable: API-sourced standardised daily market prices. (add credible point)
- Large Sample: Thousands of observations.

METHODS OF ANALYSIS

Data Preparation

Initial Model

Model Selection

Diagnostics

Model Refinement

Final Model

Research Question Answered

Clean dataset

Compute log & lagged returns

Fit full model

Check for multicollinearity (VIF)

Stepwise AIC/BIC & Remove weak predictors

Residual checks & Normality

Leverage & Cook's Distance

Re-test model assumption

Select final predictors

Interpret coefficients

MODEL AND RESULTS

$$\text{silver_ret}_t = \beta_0 + \beta_1 \text{gold_ret}_t + \beta_2 \text{sp500_ret}_t + \beta_3 \text{nasdaq_ret}_t + \beta_4 \text{oil_ret}_t + \beta_5 \text{silver_ret_lag1}_t + \beta_6 \text{gold_ret_lag1}_t + \beta_7 \text{sp500_ret_lag1}_t + \beta_8 \text{platinum}_t + \beta_9 \text{palladium}_t + \beta_{10} \text{usd_chf}_t + \beta_{11} \text{eur_usd}_t + \beta_{12} \text{sp500.spread}_t + \varepsilon_t$$

Adjusted R-Squared

	Adj R-Squared
Reduced Model	0.6608564
Full Model	0.6612

Partial F-Test

	F	p-value
Full Model	1.7438	0.1211

Reject null (weak predictors in full model) $p < 0.05$

Removed weak predictors with moderate multicollinearity (VIF):

- "silver_ret_lag1"
- "palladium"
- "usd_chf"
- "eur_usd"
- "sp500.spread"

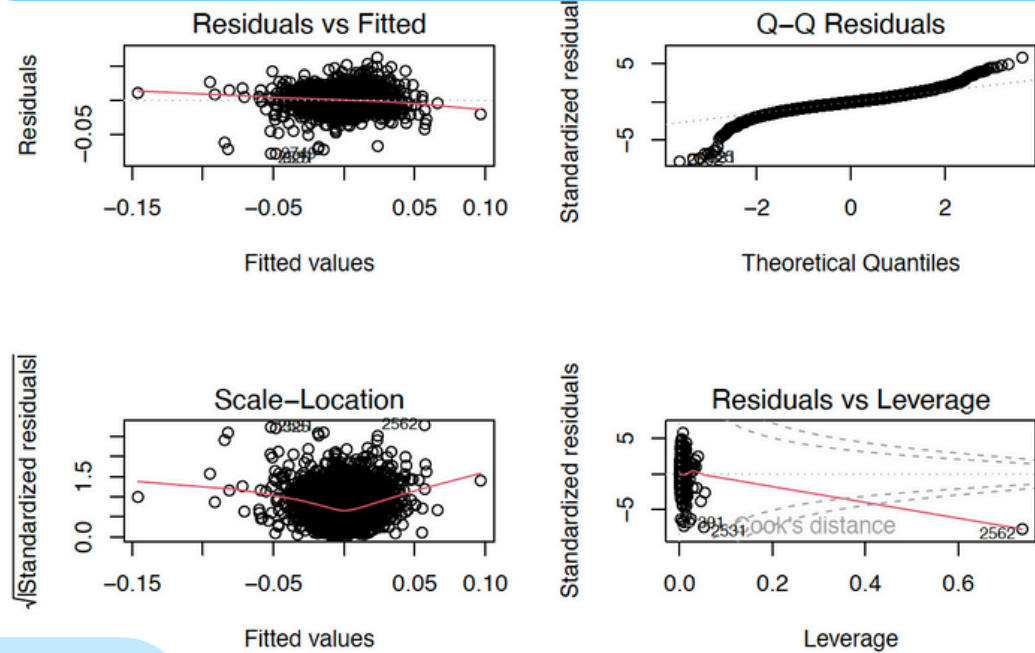
Mild heteroskedasticity

Heavy tails

Slight variance increase

No influential points

Residual Checks (Full Model)

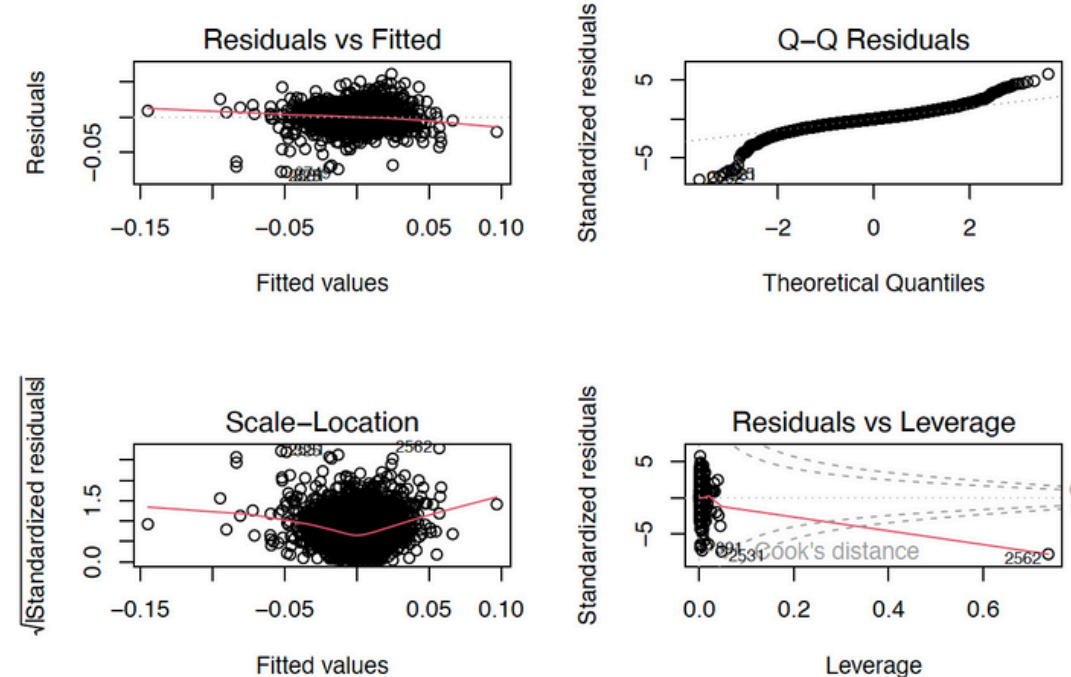


AIC/BIC Comparison

	AIC	BIC
Reduced Model	-22869.73	-22813.84
Full Model	-22868.47	-22781.54

Both AIC & BIC are lower for the reduced model. Shows better balance between model fit & model complexity

Residual Checks (Reduced Model)



Removing weak predictors preserves full-model behaviour (similar VIF)

Regression Output

Predictor	Estimate	Std. Error	p-value
Gold Return	1.4653	0.01819666	$< 2e-16$
S&P500 Return	0.3758	0.0413878	2.4×10^{-17}
Oil Return	0.002	0.000436393	4.96×10^{-6}

FINAL MODEL

- Gold: strongest positive predictor
- Oil: positive predictor
- S&P 500: negative predictor
- All signs consistent with economic theory.

LIMITATIONS

- Heavy-Tailed Residuals
- Mild Heteroskadisticity
- Omitted Variables
- Time-Series Features Not Modeled
- Market Microstructure Noise

References

- Adamkovičová, Alexandra, and Josef Blažek. 2021. "Gold and Silver as Safe Havens." SHS Web of Conferences 129: 03001.
- Cohen, Gil. 2022. "Algorithmic Strategies for Precious Metals Price Forecasting." ResearchGate.
- Fatima, Saima, Christopher Gan, and Jieqiu Hu. 2022. "Volatility Analysis of Precious Metals." Journal of Risk and Financial Management 15(10): 465.
- FranciscoGCC. 2023. "Financial Data." Kaggle.

CONCLUSION

Our results show clear evidence that silver's daily movements are jointly driven by safe-haven dynamics and procyclical forces. Gold remains the dominant predictor at about 1.47, while meaningful effects from the S&P 500 at about 0.38 and oil at about 0.02 highlight silver's sensitivity to equity sentiment and industrial demand. Overall, the evidence strongly supports the conclusion that silver behaves as a hybrid asset shaped by both defensive and growth-linked market conditions.

SUGGESTIONS

- Use Robust Standard Errors
- Controlling For Influential Observations
- Use Proxy Variables
- Use GARCH Model